

Exposé XII (continued and completed)

Proposition 7.2. Let X be a smooth scheme of dimension 1, let F be constructible, and let

$$(u, f) : (X, F) \longrightarrow (X, F)$$

be an endomorphism such that the morphism of sheaves

$$u : f^*(F) \longrightarrow F$$

is an isomorphism. Under these conditions there exist U, U', G, f', φ satisfying the conditions of 5.1, 6.1, 6.2.

Proof. Let U be the largest open subset of X on which F is locally constant, and let Y denote its complement. The set Y is stable under f :

$$f(Y) \subset Y.$$

Indeed, suppose by contradiction that there is a $y \in Y$ such that $f(y) \in U$. The sheaf $f^*(F)$ is then locally constant in a neighbourhood of y , and using the isomorphism u it follows that F is locally constant in a neighbourhood of y , which is absurd.

If f is constant, the proof of the proposition is trivial. We shall therefore suppose that f is not constant. If η is the generic point of X , then $f(\eta) = \eta$.

Let $i_\eta : \text{Spec}(k(\eta)) \hookrightarrow X$ be the canonical inclusion, put $F_\eta = i_\eta^*(F)$, and let

$$u_\eta : f_\eta^*(F_\eta) \longrightarrow F_\eta$$

be the morphism, in fact the isomorphism, of sheaves induced by u , where f_η is of course the restriction of f to $\text{Spec}(k(\eta))$.

Translating everything into terms of H -modules,

$$H = \text{Gal}(k(\eta)_s/k(\eta)), \quad k(\eta)_s \text{ a separable closure of } k(\eta),$$

one obtains the isomorphisms

$$H \xrightarrow{\Psi} H \quad F_\eta \xrightarrow{u_\eta} F_\eta,$$

where F_η has an H -module structure and the following relation holds:

$$u_\eta(gx) = \Psi(g)u_\eta x, \quad g \in H, \quad x \in F_\eta.$$

If H_1 is the largest distinguished subgroup of H of finite index such that $F_\eta^{H_1} = F_\eta$, then $\Psi(H_1) = H_1$. Such a subgroup exists because the sheaf F_η is constructible.

Let $K \supset k(\eta)$ be the Galois extension of $k(\eta)$ corresponding to the subgroup H_1 , and let X' be the normalization of X in K . It is clear that one can define an endomorphism

$$f' : X' \longrightarrow X'$$

and an endomorphism

$$H/H_1 = G \xrightarrow{\varphi} G$$

such that the system U, U', f', G, φ satisfies the required conditions.

Example 7.3. Let X be a prescheme defined over the finite field $\mathbb{F}_p$, where p is a prime number, let $\text{fr}_X : X \rightarrow X$ be the Frobenius endomorphism (cf. XIV), and let F be a sheaf of Λ -modules on X .

If Y, S are schemes over $\mathbb{F}_p$, and if Y is a scheme over S , we denote by

$$\text{Fr}_{Y/S} : Y \longrightarrow Y^{(p/S)}$$

the relative Frobenius morphism (cf. XIV 2).

We obtain a morphism of sheaves on $X_{\text{ét}}$

$$(\text{fr}_X)_*(F) \longrightarrow F$$

by setting, for every étale X -prescheme U ,

$$(\text{fr}_X)_*(F)(U) = F(\text{fr}_X^{-1}(U)) = F(U^{(p/X)}) \xrightarrow{F(\text{Fr}_{U/X})} F(U).$$

This morphism is in fact an isomorphism. Its inverse

$$F(\text{Fr}_{/X})^{-1} : F \longrightarrow (\text{fr}_X)_*(F)$$

defines by adjunction a morphism

$$\text{Fr}_{F/X}^* : (\text{fr}_X)^*(F) \longrightarrow F$$

which is an isomorphism.

If in particular X is smooth of dimension 1 and F is constructible, the pair

$$(\text{Fr}_{F/X}^*, \text{fr}_X) : (X, F) \longrightarrow (X, F)$$

therefore falls under Proposition 7.2.

[1] Wecken, F., Fixpunktklassen I, Math. Ann. 117 (1941), p. 659–671; Fixpunktklassen II, III, Math. Ann. 118 (1942–1943), p. 216–234 and p. 544–577.

Exposé XV

FROBENIUS MORPHISM AND RATIONALITY OF L-FUNCTIONS

by C. Houzel

§ 1. Frobenius morphism

No. 1. Frobenius endomorphism

Definition 1. Let p be a prime number. A prescheme X is said to be of characteristic p if $p \cdot 1_{\mathcal{O}_X} = 0$.

This condition is equivalent to the following one: the unique morphism $X \rightarrow \operatorname{Spec}(\mathbb{Z})$ factors through $\operatorname{Spec}(\mathbb{F}_p) \rightarrow \operatorname{Spec}(\mathbb{Z})$; moreover, such a factorization is unique.

We shall denote by $(\operatorname{Sch})_p$ the full subcategory of the category of preschemes whose objects are the preschemes of characteristic p . By the preceding remark, the category $(\operatorname{Sch})_p$ is naturally isomorphic to the category of $\mathbb{F}_p$ -preschemes.

If X is a prescheme of characteristic p , the pair

$$\operatorname{fr}_X = (\operatorname{id}_{|X|}, \psi), \quad \psi : \mathcal{O}_X \longrightarrow \mathcal{O}_X, \quad \psi(f) = f^p$$

for every section f of $\mathcal{O}_X$ over an open set, is an endomorphism of X , because ψ is visibly an endomorphism of the sheaf of rings $\mathcal{O}_X$.

Definition 2. Let X be a prescheme of characteristic p . The endomorphism fr_X thus defined is called the Frobenius endomorphism of X .

The Frobenius endomorphism fr_X depends functorially on the prescheme X of characteristic p ; that is, for every morphism $g : X \rightarrow Y$ of preschemes of characteristic p , the following diagram is commutative:

$$\begin{array}{ccc} X & \xrightarrow{\operatorname{fr}_X} & X \\ g \downarrow & & \downarrow g \\ Y & \xrightarrow{\operatorname{fr}_Y} & Y. \end{array}$$

In other words, we have defined an endomorphism fr of the identity functor $\operatorname{id}_{(\operatorname{Sch})_p}$.

No. 2. Relative Frobenius morphism

Let S be a prescheme of characteristic p ; all S -preschemes are then also of characteristic p . If X is an S -prescheme, one sees that fr_X factors through the projection

$$\pi_{X/S} : X \times_S (S, \operatorname{fr}_S) \longrightarrow X,$$

where (S, fr_S) denotes S regarded as an S -prescheme by means of the morphism fr_S . We shall denote the fibre product $X \times_S (S, \operatorname{fr}_S)$ by $X^{(p/S)}$, or even by $X^{(p)}$ if no confusion can result. The unique S -morphism from X to $X^{(p/S)}$ whose composite with $\pi_{X/S}$ is fr_X is denoted by $\operatorname{Fr}_{X/S}$; thus one obtains a commutative diagram in which the square is cartesian.

It is clear that $X^{(p/S)}$ depends functorially on the S -prescheme X ; the functor

$$p_S : X \longmapsto X^{(p/S)}$$

is simply base change by the morphism $\mathrm{fr}_S : S \rightarrow S$. Like every base change, it commutes with projective limits and with sums, and preserves properties such as being of finite type, separated, affine, projective, flat, étale, and so on. Commutation with projective limits shows that if X is endowed with an algebraic structure, for example a group or algebra structure, then $X^{(p/S)}$ is naturally endowed with an algebraic structure of the same kind.

Using the functoriality of fr_X with respect to X , one sees that the morphism $\mathrm{Fr}_{X/S}$ depends functorially on the S -prescheme X , so that a functorial morphism

$$\mathrm{Fr}_{X/S} : \mathrm{id}_{(\mathrm{Sch})/S} \longrightarrow p_S$$

has been defined.

Definition 3. Let S be a prescheme of characteristic p and X an S -prescheme. The S -morphism

$$\mathrm{Fr}_{X/S} : X \longrightarrow X^{(p/S)}$$

is called the Frobenius morphism of X relative to S .

The first formal properties of the Frobenius morphism are stated in the following proposition.

Proposition 1.

- a) (*Transitivity.*) Let S be a prescheme of characteristic p , and let $g : X \rightarrow Y$ be an S -morphism of S -preschemes. The natural diagram relating X , $X^{(p/S)}$, $X^{(p/Y)}$, Y , $Y^{(p/S)}$, and S is commutative and its three squares are cartesian.
- b) (*Base change.*) Let $g : S' \rightarrow S$ be a morphism of preschemes of characteristic p , and let X be an S -prescheme. There is a canonical isomorphism

$$X^{(p/S)} \times_S S' \simeq (X \times_S S')^{(p/S')}$$

functorial in X , that is, $(X^{(p/S)})' \simeq X'^{(p/S')}$ when the result of base change by g is marked with a prime; moreover

$$\mathrm{Fr}_{X'/S'} = (\mathrm{Fr}_{X/S})'$$

modulo the identification allowed by the preceding isomorphism.

The proof is immediate.

Proposition 2. Let S be a prescheme of characteristic p and X an S -prescheme.

- a) The relative Frobenius morphism

$$\mathrm{Fr}_{X/S} : X \longrightarrow X^{(p/S)}$$

is integral, surjective, and radicial; hence it is a universal homeomorphism.

- b) If $X \rightarrow S$ is an open immersion, then $X^{(p)} = X$ and $\mathrm{Fr}_{X/S} = \mathrm{id}_X$. If $X \rightarrow S$ is a closed immersion, identifying X with a closed ringed subspace $(|X|, \mathcal{O}_S/\mathcal{I})$ of S , then

$$X^{(p)} = (|X|, \mathcal{O}_S/\mathcal{I}^p)$$

and $\mathrm{Fr}_{X/S} = (\mathrm{id}_{|X|}, \varphi)$, where $\varphi : \mathcal{O}_S/\mathcal{I}^p \rightarrow \mathcal{O}_S/\mathcal{I}$ is the evident augmentation.

c) Suppose that X is locally of finite presentation over S .

1) The following three conditions are equivalent:

- (i) X is unramified over S ,
- (ii) $\mathrm{Fr}_{X/S}$ is unramified,
- (iii) $\mathrm{Fr}_{X/S}$ is a monomorphism.

2) Likewise, the following three conditions are equivalent:

- (i') X is étale over S ,
- (ii') $\mathrm{Fr}_{X/S}$ is étale,
- (iii') $\mathrm{Fr}_{X/S}$ is an isomorphism.

We prove a). The composite $\pi_{X/S} \circ \mathrm{Fr}_{X/S} = \mathrm{fr}_X$ is visibly integral, surjective, and radicial; it follows that $\mathrm{Fr}_{X/S}$ is radicial (EGA I 3.5.6 (ii)). Moreover $\pi_{X/S}$ is separated and radicial, because it is deduced from fr_S by the base change $X \rightarrow S$; it follows that $\mathrm{Fr}_{X/S}$ is integral (EGA II 6.1.5 (v)) and surjective.

The proof of b) is evident, observing that

$$\mathcal{JO}_{(S, \mathrm{fr}_S)} = \mathcal{J}^p$$

for every ideal $\mathcal{J}$ of $\mathcal{O}_S$.

Proof of c). It is known that for a morphism to be a monomorphism, respectively an isomorphism, it is necessary and sufficient that it be radicial and unramified, respectively radicial, étale, and surjective. This gives the equivalence of (ii) and (iii), respectively of (ii') and (iii'). It remains to prove that (i) is equivalent to (ii), and (i') to (ii').

For this, one uses the differential criterion of unramifiedness (SGA I 3) and the isomorphism

$$\Omega_{X/S}^1 \simeq \Omega_{X/X^{(p)}}^1$$

given by the canonical exact sequence

$$\mathrm{Fr}_{X/S}^*(\Omega_{X^{(p)}/S}^1) \longrightarrow \Omega_{X/S}^1 \longrightarrow \Omega_{X/X^{(p)}}^1 \longrightarrow 0$$

(SGA II 4), in which the first arrow is visibly zero, because $\Omega_{X^{(p)}/S}^1$ is generated by the differentials of sections of $\mathcal{O}_{X^{(p)}}$, and the differential $d_{X/S}$ vanishes on $(\mathcal{O}_X)^p$ and on $\mathcal{O}_S$. Thus (i) is equivalent to (ii).

Now suppose X is étale over S ; the same is true of $X^{(p)}$ by base change, and consequently the morphism $\mathrm{Fr}_{X/S} : X \rightarrow X^{(p)}$ is étale. Conversely, suppose that $\mathrm{Fr}_{X/S}$ is étale; it is unramified, hence X is unramified over S . In other words, the structural morphism $g : X \rightarrow S$ is locally on X , for the étale topology, a closed immersion (SGA I, cor. 7.8). We may therefore suppose that g is a closed immersion of finite presentation, and it remains to see that, if $\mathrm{Fr}_{X/S}$ is an isomorphism, then g is an open immersion. Now X is defined by an ideal $\mathcal{J}$ of finite type,

$$X = (|X|, \mathcal{O}_S/\mathcal{J}),$$

and, by b),

$$X^{(p)} = (|X|, \mathcal{O}_S/\mathcal{J}^p), \quad \mathrm{Fr}_{X/S} = (\mathrm{id}_{|X|}, \mathcal{O}_S/\mathcal{J}^p \rightarrow \mathcal{O}_S/\mathcal{J}).$$

To say that $\mathrm{Fr}_{X/S}$ is an isomorphism is therefore to say that $\mathcal{J} = \mathcal{J}^p$; since $\mathcal{J}$ is of finite type, Nakayama's lemma implies that $\mathcal{J}_s = 0$, or $\mathcal{J}_s = \mathcal{O}_{S,s}$, for every point $s \in S$, whence the conclusion.

No. 3. Effect on a module. Examples

Let S be a prescheme of characteristic p and X an S -prescheme. For every $\mathcal{O}_X$ -Module $\mathcal{E}$, a sheaf for the Zariski topology, set

$$\mathcal{E}^{(p/S)} = \pi_{X/S}^*(\mathcal{E}) = \mathcal{E} \otimes_{\mathcal{O}_X} \mathcal{O}_{X^{(p)}} = \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{O}_{(S, \text{fr}_S)}.$$

This is an $\mathcal{O}_{X^{(p)}}$ -Module, and one has

$$\mathcal{E}^{(p/X)} \simeq \text{Fr}_{X/S}^*(\mathcal{E}^{(p/S)}).$$

The most interesting case is the case $X = S$; one may then write $\mathcal{E}^{(p)}$ instead of $\mathcal{E}^{(p/S)}$ without risk of confusion, and $\mathcal{E}^{(p)} = \text{fr}_S^*(\mathcal{E})$. The adjunction morphism

$$\mathcal{E} \longrightarrow \text{fr}_{S*} \text{fr}_S^*(\mathcal{E})$$

is a morphism of abelian sheaves $\mathcal{E} \rightarrow \mathcal{E}^{(p)}$ which is not $\mathcal{O}_S$ -linear, but is “ p -linear” in the evident sense; it is visibly a universal p -linear morphism from $\mathcal{E}$ to an $\mathcal{O}_S$ -Module.

It is clear that $\mathcal{E}^{(p)}$ depends functorially on $\mathcal{E}$, and that the functor $\mathcal{E} \mapsto \mathcal{E}^{(p)}$ is right exact, but not exact in general; it commutes with tensor products:

$$(\mathcal{E} \otimes_{\mathcal{O}_X} \mathcal{F})^{(p)} \simeq \mathcal{E}^{(p)} \otimes_{\mathcal{O}_{X^{(p)}}} \mathcal{F}^{(p)},$$

and preserves properties such as being quasi-coherent, of finite type, of finite presentation, flat, locally free, and so on. In particular, if $\mathcal{L}$ is an invertible sheaf on S , one immediately sees that $\mathcal{L}^{(p)}$ is canonically isomorphic to $\mathcal{L}^{\otimes p}$, because the morphism $\mathcal{L} \rightarrow \mathcal{L}^{\otimes p}$ defined by raising sections to their p -th tensor power is a universal p -linear morphism.

If $\mathcal{A}$ is an $\mathcal{O}_S$ -Algebra, then $\mathcal{A}^{(p)}$ has an $\mathcal{O}_S$ -Algebra structure deduced from that of $\mathcal{A}$. Suppose $\mathcal{A}$ is quasi-coherent, respectively graded and quasi-coherent; the same is true of $\mathcal{A}^{(p)}$, and

$$\text{Spec}(\mathcal{A}^{(p)}) = \text{Spec}(\mathcal{A} \otimes_{\mathcal{O}_S} \mathcal{O}_{(S, \text{fr}_S)}) \simeq \text{Spec}(\mathcal{A}) \times_S (S, \text{fr}_S) = \text{Spec}(\mathcal{A})^{(p)},$$

respectively

$$\text{Proj}(\mathcal{A}^{(p)}) = \text{Proj}(\mathcal{A} \otimes_{\mathcal{O}_S} \mathcal{O}_{(S, \text{fr}_S)}) \simeq \text{Proj}(\mathcal{A}) \times_S (S, \text{fr}_S) = \text{Proj}(\mathcal{A})^{(p)}.$$

The projection $\pi_{\text{Spec}(\mathcal{A})/S}$, respectively $\pi_{\text{Proj}(\mathcal{A})/S}$, is identified with the morphism deduced from $\mathcal{A} \rightarrow \mathcal{A}^{(p)}$. On the other hand, $\text{Fr}_{\text{Spec}(\mathcal{A})/S}$, respectively $\text{Fr}_{\text{Proj}(\mathcal{A})/S}$, comes from a homomorphism of $\mathcal{O}_S$ -Algebras

$$\mathcal{A}^{(p)} = \mathcal{A} \otimes_{\mathcal{O}_S} \mathcal{O}_{(S, \text{fr}_S)} \longrightarrow \mathcal{A}$$

which sends $a \otimes f$ to $a^p f$, for every section a of $\mathcal{A}$ and every section f of $\mathcal{O}_S$. To see this, observe that the composite of this homomorphism with $\mathcal{A} \rightarrow \mathcal{A}^{(p)}$, on the right, is raising to the p -th power in $\mathcal{A}$.

Consider the particular case where $\mathcal{A}$ is the symmetric algebra $\text{Sym}(\mathcal{E})$ of a quasi-coherent $\mathcal{O}_S$ -Module $\mathcal{E}$; it is clear that $\mathcal{A}^{(p)} \simeq \text{Sym}(\mathcal{E}^{(p)})$, hence

$$\mathbb{V}(\mathcal{E}^{(p)}) \simeq \text{Spec}(\mathcal{A}^{(p)}) \simeq \text{Spec}(\mathcal{A})^{(p)} = \mathbb{V}(\mathcal{E})^{(p)},$$

$$\mathbb{P}(\mathcal{E}^{(p)}) \simeq \text{Proj}(\mathcal{A}^{(p)}) \simeq \text{Proj}(\mathcal{A})^{(p)} = \mathbb{P}(\mathcal{E})^{(p)}.$$

As for $\mathrm{Fr}_{\mathbb{V}(\mathcal{E})/S}$, respectively $\mathrm{Fr}_{\mathbb{P}(\mathcal{E})/S}$, it factors in the following way:

$$\mathbb{V}(\mathcal{E}) \longrightarrow \mathbb{V}(\mathrm{Sym}^p(\mathcal{E})) \longrightarrow \mathbb{V}(\mathcal{E}^{(p)}),$$

respectively

$$\mathbb{P}(\mathcal{E}) \longrightarrow \mathbb{P}(\mathrm{Sym}^p(\mathcal{E})) \longrightarrow \mathbb{P}(\mathcal{E}^{(p)}),$$

where the first arrow is defined by the canonical morphism

$$\mathrm{Sym}(\mathrm{Sym}^p(\mathcal{E})) \longrightarrow \mathrm{Sym}(\mathcal{E})$$

which comes from $\mathrm{Sym}^p(\mathcal{E}) \rightarrow \mathrm{Sym}(\mathcal{E})$, while the second is defined by the morphism of $\mathcal{O}_S$ -Modules

$$\mathcal{E}^{(p)} \longrightarrow \mathrm{Sym}^p(\mathcal{E})$$

coming from the p -linear morphism $\mathcal{E} \rightarrow \mathrm{Sym}^p(\mathcal{E})$ given by raising sections to their p -th symmetric power. In the case $X = \mathbb{P}(\mathcal{E})$, let us compute the fundamental sheaf $\mathcal{O}_{X^{(p)}}(1)$:

$$\mathcal{O}_{X^{(p)}}(1) = (\mathrm{Sym}(\mathcal{E})^{(p)}(1))^\sim = (\mathrm{Sym}(\mathcal{E})(1)^{(p)})^\sim \simeq \mathcal{O}_X(1)^{(p/S)}.$$

When $\mathcal{E} = \mathcal{O}_S^n$, $\mathrm{Sym}(\mathcal{E}) = \mathcal{O}_S[T_1, \dots, T_n]$ and $X = \mathbb{V}(\mathcal{E}) = S[T_1, \dots, T_n]$, respectively $X = \mathbb{P}(\mathcal{E}) = \mathbb{P}_S^{n-1}$, it is clear that $\mathcal{E}^{(p)} \simeq \mathcal{O}_S^n = \mathcal{E}$, so that $X^{(p)} = X$. The Frobenius morphism $\mathrm{Fr}_{X/S}$ comes from the endomorphism of the algebra $\mathcal{O}_S[T_1, \dots, T_n]$ which sends T_i to T_i^p ($i = 1, \dots, n$).

Now consider a prescheme X , affine of finite type, respectively projective, over S , defined by an ideal, respectively a homogeneous ideal,

$$\mathcal{J} \subset \mathcal{O}_S[T_1, \dots, T_n].$$

Thus $X = \mathrm{Spec}(\mathcal{A})$, respectively $X = \mathrm{Proj}(\mathcal{A})$, where

$$\mathcal{A} = \mathcal{O}_S[T_1, \dots, T_n]/\mathcal{J}.$$

One sees that

$$\mathcal{A}^{(p)} \simeq \mathcal{O}_S[T_1, \dots, T_n]/\mathcal{J}^{(p)},$$

where $\mathcal{J}^{(p)}$ is the image in $\mathcal{O}_S[T_1, \dots, T_n]$ of $\mathcal{J} \otimes_{\mathcal{O}_S} \mathcal{O}_{(S, \mathrm{fr}_S)}$; the ideal $\mathcal{J}^{(p)}$ is deduced from $\mathcal{J}$ by raising the coefficients of the polynomials to the p -th power. One has $X^{(p)} = \mathrm{Spec}(\mathcal{A}^{(p)})$, respectively $\mathrm{Proj}(\mathcal{A}^{(p)})$, and the Frobenius morphism

$$\mathrm{Fr}_{X/S} : X \longrightarrow X^{(p)}$$

is defined by raising the coordinates T_i to the p -th power; this determines a morphism $\mathcal{A}^{(p)} \rightarrow \mathcal{A}$.